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Ridge Regression

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Principle Component Analysis (PCA)

Video: Principle Component Analysis (PCA) Overview
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Principle Component Analysis

Principal Component Analysis (PCA) is a dimensionality reduction technique used to transform a high-dimensional dataset into a lower-dimensional space while preserving the most important patterns and variations in the data. It achieves this by identifying and extracting the principal components, which are new uncorrelated variables that are linear combinations of the original variables.

The goal of PCA is to find a set of orthogonal axes (principal components) in the feature space that capture the maximum amount of variance in the data. The first principal component accounts for the largest possible variance, the second component accounts for the second largest variance, and so on. Each subsequent component captures decreasing amounts of variance.

The steps involved in performing PCA are as follows:

1. **Standardize the data:** PCA is sensitive to the scale of the variables, so it is necessary to standardize the variables to have zero mean and unit variance.
2. **Compute the covariance matrix or correlation matrix:** Depending on the nature of the data and the objective of the analysis, either the covariance matrix or the correlation matrix is calculated. The covariance matrix measures the linear relationship between variables, while the correlation matrix measures the linear relationship and scale independence.
3. **Calculate the eigenvectors and eigenvalues:** The eigenvectors represent the principal components, and the eigenvalues indicate the amount of variance explained by each principal component. The eigenvectors and eigenvalues are derived from the covariance or correlation matrix.
4. **Select the principal components:** The principal components are selected based on the eigenvalues. Typically, the components associated with the largest eigenvalues are chosen, as they explain the most variance in the data. The number of components selected determines the dimensionality of the reduced feature space.
5. **Transform the data:** The original data is projected onto the selected principal components to obtain the new lower-dimensional representation of the data.

PCA has various applications, including:

- **Dimensionality reduction:** PCA helps in reducing the number of variables in a dataset while preserving the important information, which can be beneficial for visualization, computational efficiency, and avoiding the curse of dimensionality.
- **Data visualization:** PCA can be used to visualize high-dimensional data by plotting the data points along the principal components.
- **Data preprocessing:** PCA can be used as a preprocessing step to remove noise, redundancy, or multicollinearity in the data before applying other machine learning algorithms.
- **Feature extraction:** The principal components can serve as new features that capture the most significant patterns and variations in the data, potentially improving the performance of subsequent machine learning models.

Overall, PCA is a valuable tool for exploratory data analysis, feature engineering, and dimensionality reduction in various domains, such as image processing, finance, genetics, and natural language processing.

The following link goes into more details and explains the concepts.

[A Step-by-Step Explanation of Principal Component Analysis \(PCA\).](#)

